

Project Proposal

Post-Surgery Recovery Assistant

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Project Description: Post-Surgery Recovery Assistant is a virtual assistant aimed at offering specific and relevant information concerning the period after surgery. Some of questions that will be resolved by the assistant are: Recovery; Should we take pain-killing drugs; Physical therapy; Diet & Nutrition. Taking advantage of natural language processing and retrieval-augmented generation (RAG) the assistant will provide the patient with only the needed information concerning particular type of surgery and its aftermath.

Use Case: After surgery, many patients are likely to have many questions about their healing process. This assistant will help in creating an organized information system for care providers and patients avoiding extra pressure on the providers' shoulders and helping every patient become proactive in their treatment.

Data Sources: As a reference material for developing the Post-Surgery Recovery Assistant, I will use some credible local and international sources of data to ensure that I get precise and rather detailed data. Clinical references will seek from organizations such as MedlinePlus, American College of Surgeons and the NHS (UK) all of which provide detailed recovery routines that are useful for patient management. Moreover, I will use the publicly available datasets, such as American College of Surgeons National Surgical Quality Improvement Program (NSQIP) that gives data on surgical results and Medicare data from government health agencies that gives data on time to recovery and results.

I also will use the patient education handouts from the UpToDate, which is a source that provides more than 8000 handouts, and Clinical Key for Nursing – this resource covers the nursing patient education issues. Moreover, open-source medical archives as, for example, PubMed Central – a free database offering the access to millions of medical articles and literature – or Cerner Health Facts, a de-identified EHR database) will be integrated to the developing knowledge base of the medical assistant.

The collected data will primarily be structured as textual datasets in the formats such as CSV, JSON and immensely convenient for manipulating and processing. Thus, documents will differ in length: short FAQs containing 1-3 sentences & longer recovery protocols of 1-2 pages. Web scraping will be implemented to obtain data from various sources and medical literature, where guidelines compiled from open datasets will make it possible for the assistant to have correct and up-to-date information pertaining to patients care after surgery.

Model Selection: Llama, more precisely the 7B version, together with BERT will be the foundational language models that I plan to employ for the assistant.

Llama will be used for the generation of responses with respect to the received patient inquiries to guarantee its relevance to the context.

In the aspect of information retrieval jobs, BERT will be applied to comprehending and, therefore, matching between text queries and documents or relevant resources.